

Examiner reports

Q1.

Part (a) was done very well by a good number of the candidates. A few worked with the magnitudes of the quantities, starting with the magnitude of the force, rather than with the actual vectors. Part (b) was found to be much more demanding. The two main errors made by the candidates were to assume that the particle still had the same acceleration as in part (a) or to use a time of 45 s rather than 25 s for the second stage of the motion.

Q2.

Parts (a) and (b) of this question were done very well by almost all of the candidates. Part (c) proved to be a little more challenging. One common error was simply to calculate the product of the mass and the acceleration. The other common error was to include the correct terms but to make a sign error.

Q3.

In part (a) of this question, some candidates used an incorrect angle when trying to resolve. Some others simply assumed that the normal reaction force was equal to the weight. In part (b), those who had had difficulties resolving in part (a) had further difficulties here. However there were a reasonable number of candidates who were able to find the correct coefficient of friction. A large number of candidates did not include an inequality in their working. In part (c), a good number of candidates calculated the magnitude of the friction force correctly, but some then went on to assume that this was the only force acting.

Q4.

Again there were a good number of complete solutions to this question. Some candidates did have difficulties with parts (c) and (d). This was generally due to ignoring the friction force and simply assuming that the tension was equal to the product of the mass and the acceleration.

Q5.

This question was generally done well. Some candidates lost marks in part (a) because they ignored the instruction "By forming an equation of motion for each particle...". A few candidates produced a negative acceleration in part (a), which was satisfactory because it was consistent with their equations, but then substituted a positive acceleration into an equation in part (b) to find an incorrect value for the tension.

There were many candidates who answered part (c) well including some of those who had had difficulty with the previous parts.

Q6.

The majority of candidates made a good attempt at this question and there were many completely correct solutions. In part (a), some of the candidates considered the three stages of the motion, but made errors when calculating one or more of these distances. In part (b), a very small number of candidates confused the numbers and gave the reciprocal of the acceleration as their answer. The majority of force diagrams in part (c) were correct,

but the most common error was to omit one force.

Calculating the force in part (c) caused the most difficulties. Some candidates made a sign error in their equations, but the most common error was to simply multiply the acceleration by the mass, without taking account of the resistance force.

Q7.

There were many good responses to this question, with a good number of candidates gaining full marks. In part (a), the most common error was where candidates did not use a constant acceleration equation but used the argument $v = 0.8/2 = 0.4$, $a = 0.4/2 = 0.2$. In part (b), the most common error was due to incorrect signs in the equation of motion. In part (c), some candidates omitted a term from the equation and some did not give the answer to two significant figures as instructed.

Q8.

Drawing the force diagram in part (a) did cause difficulties for the candidates with several wanting to include a force directed down the slope. Part (b) was done correctly by almost all of the candidates. In part (c), several candidates did not form a three term equation, either omitting the component of the weight down the slope or the friction force. Most candidates produced a good attempt at part (d).

Q9.

Part (a) was done very well, and many of the candidates seemed to have benefited from the printed answer. Part (b) however, produced a wide variety of responses. There was very little use of the constant acceleration equations in this question, which was encouraging. The main problems that the candidates encountered were with the integration. Errors such as omitting a negative sign or incorrectly splitting the $1/50 - v$ into two fractions were the most common errors.

Q10.

The first three parts of this question were generally done very well. Some candidates made arithmetic or algebraic errors, but these were not common place. In part (d), a few candidates simply calculated the product of the mass and the acceleration.

Q11.

Part (a) was done very well and many candidates were able to obtain the acceleration given in the question from correct working. Part (b) proved to be more problematic. The equation of motion was often set up incorrectly, the negative acceleration being a particular problem for some. In part (b), a number of candidates subtracted the 300 N from their previous answer rather than adding it on. Some candidates did not realise that they could simply modify their answer to part (b) and did a lot more work than was needed. There were some good answers to part (d), often expressed well. There were also a number of candidates who gave spurious answers.

Q12.

There were many good solutions to this question. There were a number of candidates who included additional forces on their force diagrams. Candidates should be encouraged

to only show the forces acting on force diagrams. Including velocities or components as well as the forces often confuses diagrams and can cause marks to be lost. Part (b) was done very well. In part (c), the most common error was only to include one force, typically the friction force, and not to include the component of the weight that acts down the slope. Two other less common errors were to use $80ga$ instead of $80a$ and to resolve the weight incorrectly or not at all.

Q13.

Part (a) of this question was generally done very well, with many correct solutions. It was good to see a large number of candidates setting up an equation of motion for each particle and then using these to obtain the acceleration. The candidates seemed familiar with the topic and knew exactly what the question required.

The candidates found part (b) very much more demanding and there were relatively few correct solutions. Some candidates did not attempt this part of the question and some others had no clear strategy for finding a solution. One error was to find the time to travel 2m and then use this time to show that they travelled 2 m. Some candidates found the speed, when the rope became slack, but could not then progress to deal with the motion that took place after this had happened.

Q14.

There were a number of good solutions to part (a). The main errors were due to missing negative signs in the differential equation, the use of $v \frac{dv}{dx}$ instead of $\frac{dv}{dt}$ or algebraic errors in the working. Very few candidates tried to use the constant acceleration equations. Part (b) was answered correctly by the vast majority of the candidates.

Q15.

This question was often done well. Parts (a) and (b) were answered correctly by a large proportion of the candidates, but part (c) caused more difficulties. The most common incorrect responses were 1225 N, caused by a sign error in the equation of motion, or 1625 N, obtained by ignoring the resistance force.

Q16.

Many of the candidates found this question more difficult than expected. Part (a) was generally done well, but problems then emerged in part (b). Here many candidates found the magnitude of the acceleration and then used this scalar quantity in part (c). It was not uncommon to see an equation that involved a velocity that was a vector and an acceleration that was a scalar. Some of the candidates who worked through correctly stopped in part (c) and did not find the distance that had been requested.

Q17.

Part (a) was generally done well. Some candidates made arithmetic errors and a small number misquoted the constant acceleration equation. The force diagrams were drawn very well. Most of the candidates found it easy to calculate the magnitude of the friction force, although a small number took the reaction force as being equal to the weight. Many candidates had difficulty finding the tension in the rope.

Q18.

The candidates generally did very well with this question. Apart from a few minor errors in earlier parts, the only problems arose in part (c)(ii), where a number of candidates found the difference between 768 and 450, instead of the sum.

Q19.

The candidates had very few problems with this question, with a high proportion of completely correct solutions.

Q20.

Part (a) was generally done very well. Very few candidates did not form two equations of motion as requested. A few had problems dealing with the friction force. Part (b) proved to be more difficult for some candidates, who dropped the negative sign and simply took the acceleration to be 1.225. Many of these candidates then just dropped a negative sign in the middle of their working. Part (c) was done well by some candidates, but again the negative acceleration caused problems for some.



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